COMPUTER NETWORKS-
DIGITAL ASSIGNMENT 3

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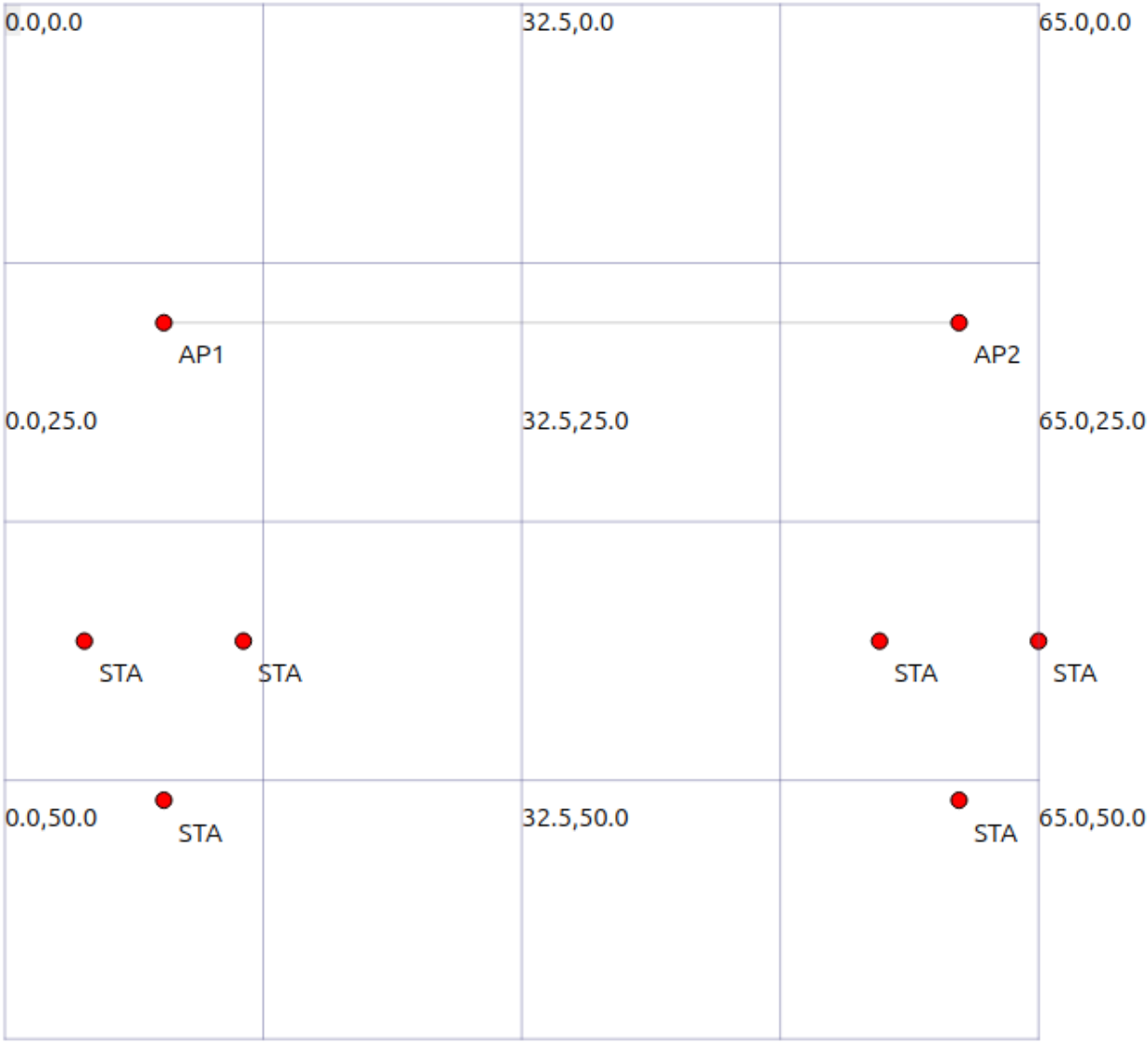
(I) Introduction to Network Simulation:

- Network simulation is applied to simulate real world communication systems in a controlled environment.
- The project is an implementation of a Smart Campus Network Simulation based on the use of ns-3 to examine the behavior of mixed traffic in semi-realistic conditions.
- The simulation includes:
    - Video streaming (UDP)
    - File transfer (TCP)
    - IoT communication (UDP)

(II) Network Design

A **Hybrid Network Topology** is implemented consisting of:

- 2 Wi-Fi Access Points (AP1, AP2)
- 6 Station Nodes (STA)
- A Point-to-Point backbone between the 2 Wi-Fi APs



- Nodes are divided into two Wi-Fi networks
- APs are connected via backbone (192.168.10.0/24)
- Subnets:
    - 10.1.1.0/24 (AP1 side)
    - 10.2.1.0/24 (AP2 side)

(III) Simulation Parameters:

Parameter	Value
Data Rate	10 Mbps
Delay	5 ms
Nodes	8
Mobility	Static
Error Model	~2% packet loss
Simulation Time	10 sec

(IV) Traffic Configuration:

Traffic Type	Protocol	Description
Video Streaming	UDP	Continuous high-rate traffic
File Transfer	TCP	Reliable data transmission
IoT Updates	UDP	Periodic low-rate packets

(V) The Aspect of “Controlled Realism”

The simulation includes:

- Packet loss using error model (~1–3%)
- Moderate congestion
- Static nodes
- Shared wireless medium

(VI) Performance Metrics

- Throughput
- Packet Delivery Ratio (PDR)
- End-to-End Delay
- Jitter
- Packet Loss

(VII) Final Output(Overall/Avg Metrics):

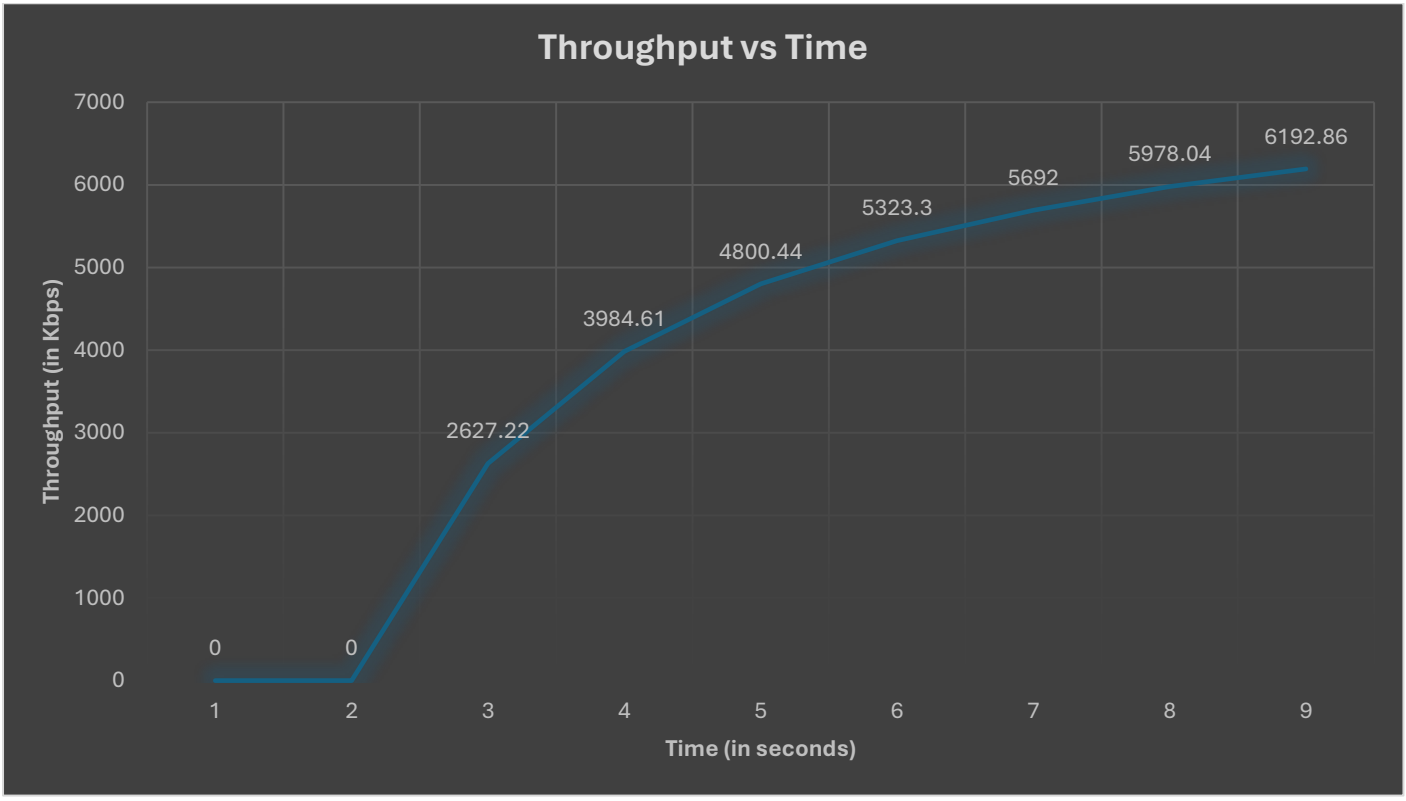
```
keshav@UbuntuOs:~/ns-3-dev$ ./ns3 run scratch/smart-campus
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable /home/keshav/ns-3-dev/build/scratch/ns3-dev-smart-campus-default
[0/2] Re-checking globbed directories...
ninja: no work to do.
Max Packets per trace file exceeded

=== FINAL METRICS ===
Throughput: 6374.01 Kbps
PDR: 98.7891 %
Delay: 78.2789 ms
```

(VIII) Performance Analysis:

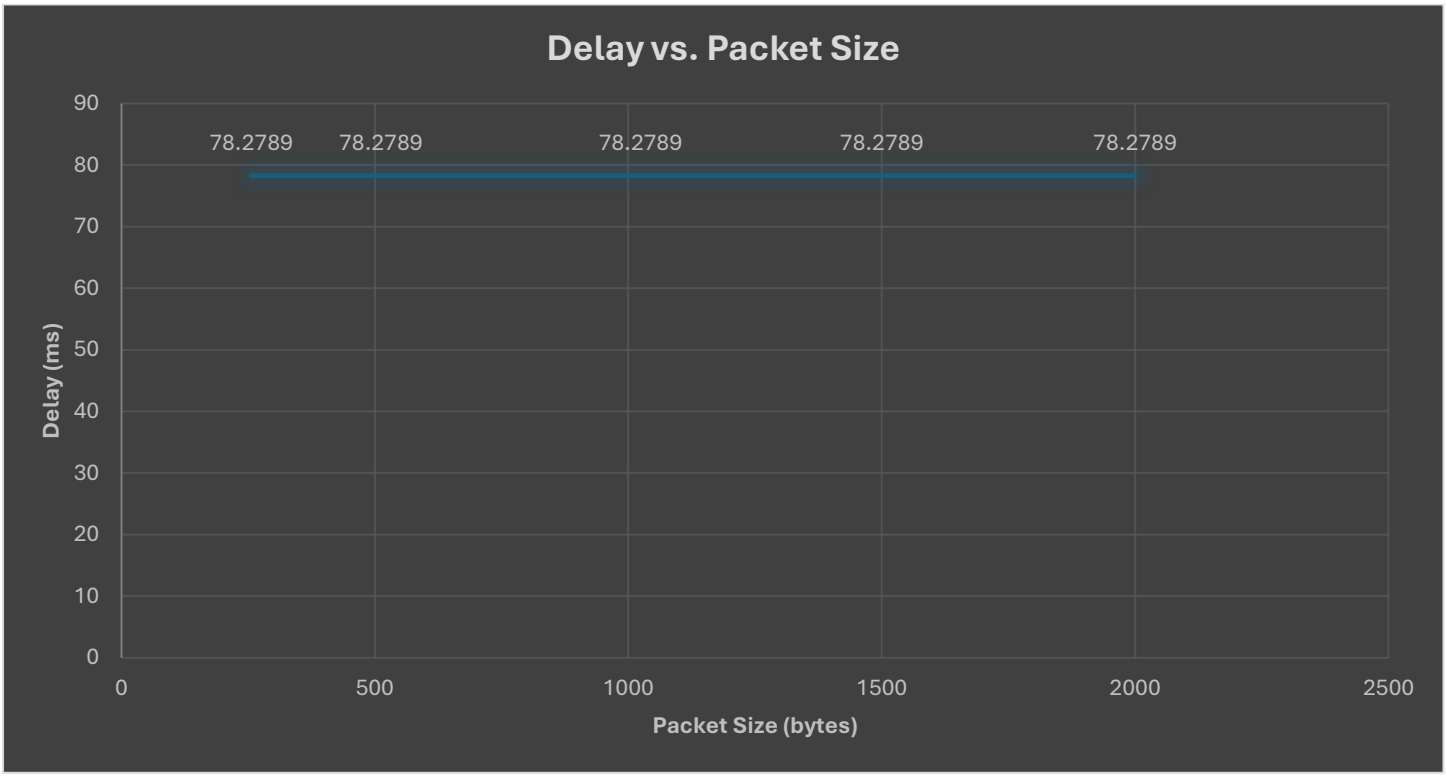
(a) Throughput vs Time (From throughput.txt log file):

Time (s)	Throughput (Kbps)
1	0
2	0
3	2627.22
4	3984.61
5	4800.44
6	5323.3
7	5692
8	5978.04
9	6192.86



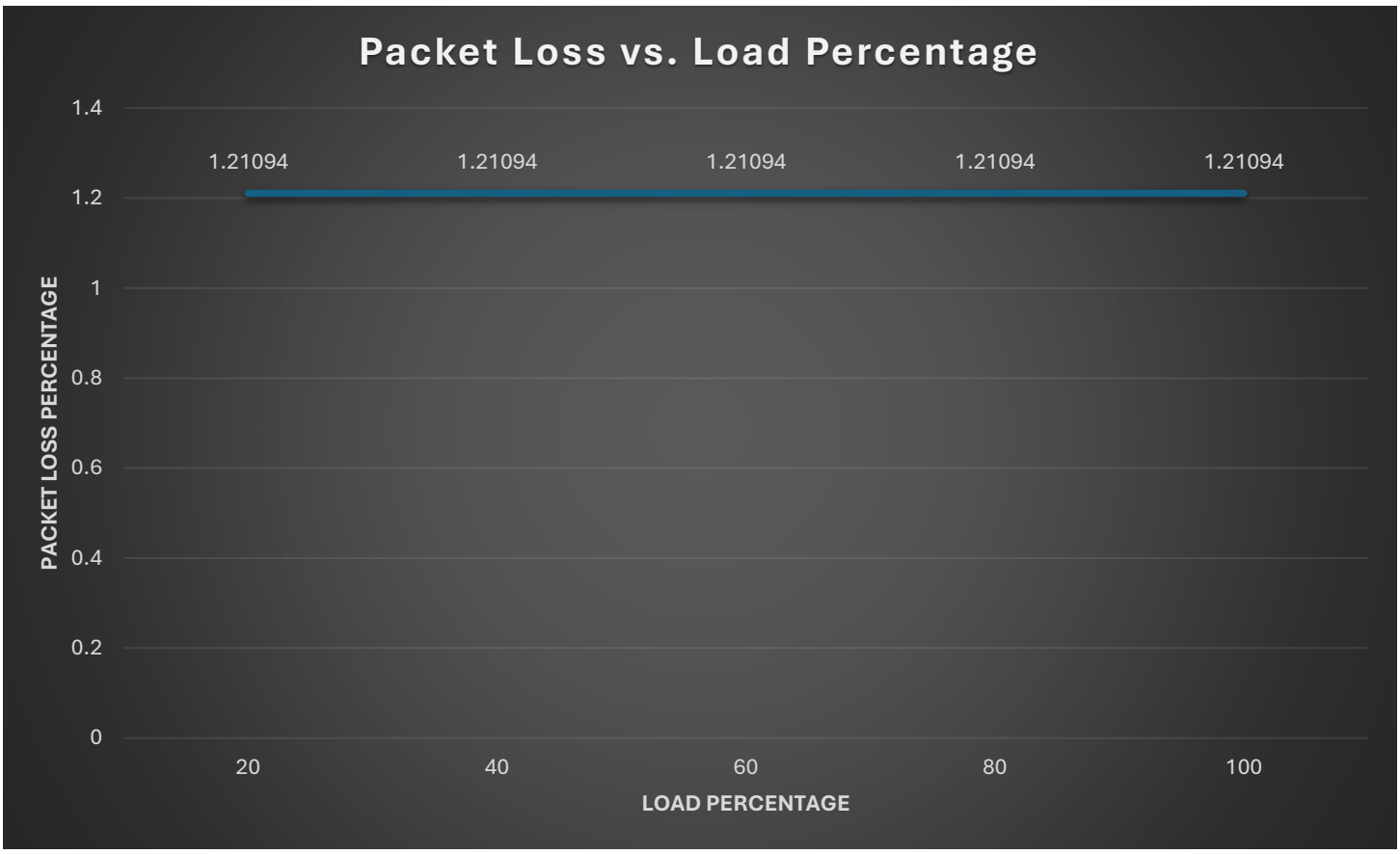
(b) Packet Size vs Delay (from delay_vs_size.txt log file)

Packet Size (bytes)	Delay (ms)
256	78.2789
512	78.2789
1024	78.2789
1500	78.2789
2000	78.2789



(c) Load vs. Packet Loss (from loss_vs_load.txt log file):

Load (%)	Packet Loss (%)
20	1.21094
40	1.21094
60	1.21094
80	1.21094
100	1.21094



(IX) Observations:

- **Throughput Behavior:**
  - Initially low (startup phase)
  - Increases steadily as flows stabilize
  - Indicates efficient network utilization

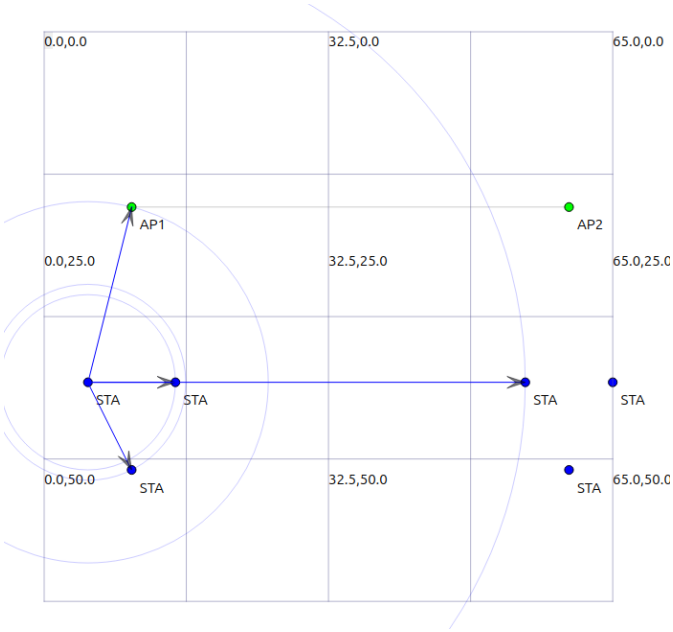
- **Delay Analysis:**
  - Delay remains constant (~78 ms)
  - Indicates stable network conditions
  - No excessive congestion

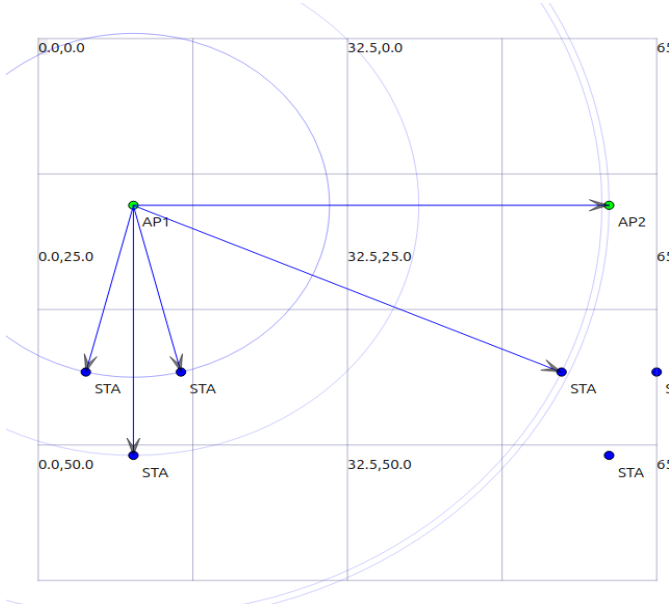
- **Packet Loss Analysis:**
  - Constant loss (~1.21%)
  - Matches configured error model
  - Shows controlled realism

TCP vs. UDP Comparison:

Aspect	TCP	UDP
Reliability	High	Low
Speed	Moderate	High
Use Case	File Transfer	Streaming, IoT

(X) NetAnim Analysis (Refer mp4 video for animation demo)





Observations:

- Continuous packet flow visible
- Multiple concurrent transmissions
- Clear separation of nodes and APs

(XI) Limitations

- No mobility (static nodes)
- Simplified channel model
- Limited interference modeling
- Same delay values across packet sizes

(XII) Conclusion

The Smart Campus Network Simulation using ns-3 successfully demonstrates:

- Mixed traffic handling
- TCP vs UDP behavior
- Impact of controlled packet loss
- Realistic network performance trends

The system provides a strong approximation of real-world campus networking conditions.